

Python Programming

Unit 06 – Lecture 01: NumPy Basics (ndarray, Operations, Broadcasting)

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February 20, 2026

Repository: <https://github.com/tali7c/Python-Programming>

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Learning Outcomes

- Explain why NumPy is used for numerical computing

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- Create arrays and understand dtype and shape

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- Create arrays and understand dtype and shape
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- Explain broadcasting with examples

Why NumPy?

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- Fast numerical computing (implemented in C)
- Compact arrays with fixed data types
- Vectorized operations (no Python loops for many tasks)
- Foundation for Pandas, ML, and scientific computing

List vs NumPy Array

- List: can store mixed types, slower for numeric math

```
import numpy as np
x = np.array([1, 2, 3])
print(x * 2)    # [2 4 6]
```

List vs NumPy Array

- List: can store mixed types, slower for numeric math
- NumPy array: fixed dtype, faster operations

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- `np.array([...])`
- `np.zeros((r,c)), np.ones((r,c))`
- `np.arange(start, stop, step)`
- `np.linspace(start, stop, num)`

Array Attributes

■ shape, ndim, size, dtype

```
a = np.array([[1, 2], [3, 4]])  
print(a.shape, a.ndim, a.size, a.dtype)
```

Broadcasting (Idea)

- NumPy can apply operations between arrays of different shapes

```
a = np.array([1, 2, 3])  
print(a + 10)    # [11 12 13]
```

Broadcasting (Idea)

- NumPy can apply operations between arrays of different shapes
- Example: add a scalar to every element

```
a = np.array([1, 2, 3])  
print(a + 10)    # [11 12 13]
```

Demo: Array Creation + Broadcasting

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Checkpoint 1

Question: Why is `[1,2,3] * 2` different from `np.array([1,2,3]) * 2`?

Checkpoint 2

Question: What does broadcasting mean in NumPy?

Think-Pair-Share

Discuss:

- When would you still use a Python list instead of a NumPy array?

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- Vectorized operations avoid explicit loops

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- NumPy arrays are fast and have fixed dtype
- Vectorized operations avoid explicit loops
- Broadcasting applies operations across compatible shapes

Exit Question

Name any two NumPy functions used to create arrays.